Bright Box

Portable Solar Charger & Light

Group 1.13: *Mahi Sheth, Sarim Javed, Kritika Dixit, Amalia Cardoso*

Meet The Team

Sarim Javed

Project Manager

Computer Science

Sarim brings experience in product development and has established himself as a professional problem solver with experience creating smart devices. He has participated in hackathons, which gives him technical expertise

Mahi Sheth

Project Manager

Computer Science

Mahi is an experienced event manager and organizer which allows her to manage the team members and keep the project on track

Kritika Dixit

Documentation Lead

Chemical Engineering

Kritika brings entrepreneurial experience to the team as she helped make an orphanage sustainable by effectively managing operations and organizing creative fundraisers

Amalia Cardoso

Researcher

Mechanical Engineering

Amalia brings hands-on technical experience as she worked to build a large competition robot, managing tasks like field calibration, wire management, proportionality components etc.,

What Makes Us Good?

#1

Diverse academic backgrounds allow us to merge science with technology to create holistic and practical solutions

#2

Defined roles, according to everyone's strengths and **effective collaboration** allows us to have maximum impact

#3

Experience with variety of **design challenges** (hackathons, robotics competitions) and **community engagement** activities gives us the required hard and soft skills

#4

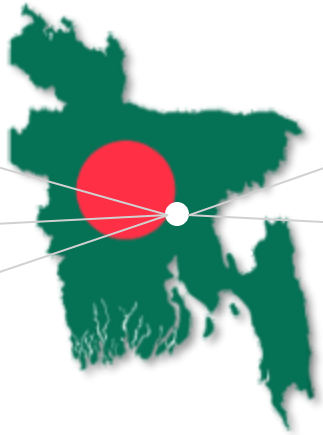
Cultural awareness and personal understanding of South Asian energy issues gives us personal insight into the problem and feasible solutions

Which Region Is Our Product For?

24 million residents (3rd Most Densely Populated City In the World)

19 % of the population are under the **poverty line**

Tropical monsoon climate: hot, humid, and flood-prone



Dhaka, Bangladesh

78% literacy rate (highest in the country)

Accounts for **35-40%** of Bangladesh's GDP

Why Dhaka?

Because there's an opportunity to create a ***real*** difference for ***millions of people***

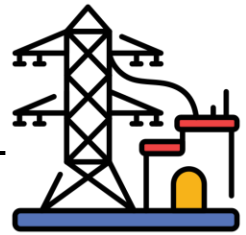
According to UN, ***all 17 SDG's*** have to be addressed in Dhaka, meaning that there's an opportunity to create a ***big impact***

With ***~5 million*** people below the poverty line, a successful solution for Dhaka would contribute massively to break the ***global poverty cycle***

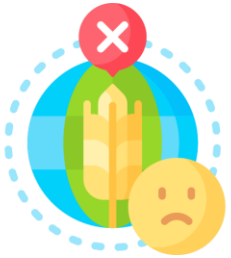
Dhaka is one of the fastest growing cities, increasing its population density even more. So, there's a need for ***optimizing the use of resources***

Identifying Dhaka's Needs

Home to 24 million people and one of the fastest growing cities in the world faces four major issues today



Electricity Shortage



Nutritious Food

Sanitation

Pollution Free Environment

We're focusing on electricity shortage. Here's why:

1. Lack of electricity impacts every **economic, social, and cultural institution** and will play a big role in boosting the economy and alleviating poverty
2. Power outages impact access to education which **decreases literacy** rates, directly increasing poverty
3. Power outages create public **safety risks** and impact communication

Setting the Success Criteria

Problem: Electricity shortage

Solution: Increase access to electricity

Criteria	Specifications
Affordability	<$25 per unit for production cost
Performance	≥5 hours of lighting, 2 phone charges/day
Sustainability	20W solar panel, recyclable casing materials
Reliability	500+ charge cycles; easy local repair
Accessibility	UX optimized for ease of use for all age groups

Relevance with Broader Context

Context:

Poverty is one of the biggest issues our world faces today. One of the ways to alleviate poverty is to provide families with more efficient ways of meeting their needs. Our product does so by improving the access to electricity for families, which is one of the contributors to multidimensional poverty.

Our Solution:

- ✓ Aligns with **UN Sustainable Development Goal 7 (Affordable and Clean Energy)**
- ✓ Supports **Goal 4 of the UN SDG** (Quality Education)
- ✓ Addresses **multidimensional poverty** by improving income potential, education, health, and safety

Our Solution: Bright Box

Problem: Electricity shortage → **Solution:** Increase access to electricity

↓

How? → **Bright Box**: a solar-powered charging device & light

- Compact and **portable** design
- 20W solar panel + rechargeable **Li-ion battery**
- Fully charges **2-3 smartphones** per day
- Provides lighting and phone charging during **outages**
- Built from **recycled**, durable materials and weather-resistant casing
- Empowers users with reliable, **renewable** energy access
- **Locally manufactured** and repairable

Comparison: Bright Box vs Others

Bright Box

- Affordable
- Dual functionality: Charging + Lighting
- Optimized for Dhaka's rain & dust conditions
- Locally manufactured and repairable
- Does not require electricity to function
- Hazard-free and easy to use

Others

- Solar lamps, power-banks, etc. are costly
- No dual functionality solutions exist today
- Current charged lights currently only work indoors
- Imported and difficult to repair
- Current powerbanks only run when pre-charged
- Traditional kerosene lamps pose hazards

Cost Analysis

Labor Cost Formula: Cost = (Personnel Hours) × (1 + 0.78) × $40

Labor Cost:

- Sarim Javed – 5 hrs → $356
- Mahi Sheth – 5 hrs → $356
- Kritika Dixit – 5 hrs → $356
- Amalia Cardoso – 5 hrs → $356
- **Total = 20 hrs → $1424 prototype cost**

Production Cost:

- **Materials:**
  - Solar panel: $10
  - Battery: $8
  - Charge Controller: $4
  - LED Light: $3
  - Casing: $2
  - Wiring, Switches, Connector: $2
- **~ $29/Unit**

~ $23-28 / unit*

*After scalability enabled by local manufacturing and bulk materials purchase

Impact of Bright Box in Dhaka

Bright box will give everyone in Dhaka access to electricity during power outages. This will have the following impact:

Economic:

- Increased working hours for small businesses and street vendors
- New market for device manufacturing and repairs will increase employment

Education:

- Increased number of hours for students to learn using electricity
- Better and longer access to the internet, enabling self learning

Health & Public Safety:

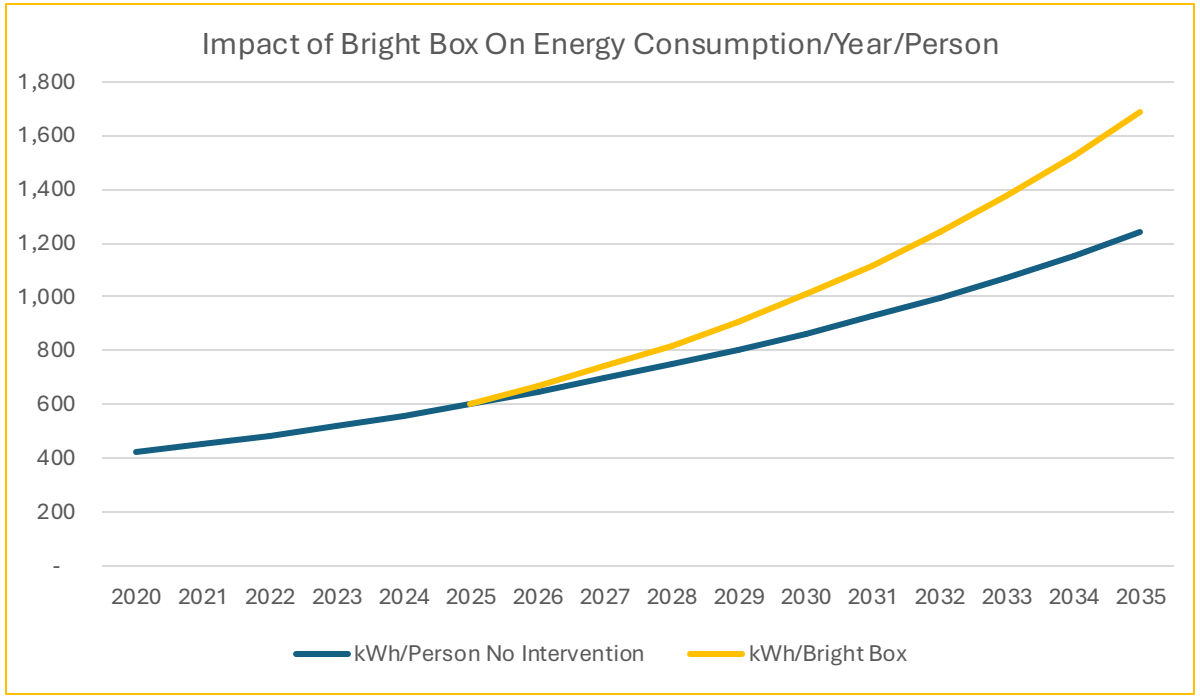
- Reduced use of kerosene lamps which expose people to harmful smoke and increase risk of fire

Environment:

- Cleaner, renewable energy will have no adverse impact on the environment
- Reduced emissions of CO2 as a result of replacing kerosene oil lamps

Impact of bright box in terms of higher energy production and lowered CO2 emissions shown through mathematical models next

Impact of Bright Box – More Electricity



- Electricity consumption per person in Bangladesh has been increasing by **7.5% per year** on average for the past 5 years
- We are assuming that it will keep increasing at this rate consistently over the next 10 years
- Bright Box Increases electricity access per person by **~0.06 kWh/day,** which is a 3% increase in annual consumption
- If Bright Box is adopted, it will increase energy consumption by **11.5%** year on year (7.5% current increase + 3% increase because of Black Box)

Thank You

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